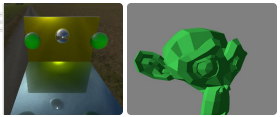
Study 1: 32 programs, checker frozen before its first run



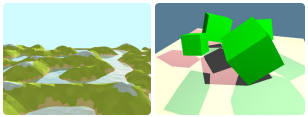
Qt Quick 3D 14 programs · Qt 6.4.2

frozen		**8%** 88 / 12 / 1,026
observed		**88%** 88 / 12 / 0
release		**8%** 88 / 10 / 1,026 / 2u

3 true causes

- ↑ acquire semaphore reused
- ↑ present semaphore reused
- ● texture visible to the vertex stage only, sampled by the fragment stage

false alarms: host writes into per-frame slots smaller than a page (conservative capture)



wgpu 15 examples

frozen		**0%** 0 / 0 / 945
observed		**0%** 0 / 0 / 906
release	no report	0

0 true causes

false alarms: three checker defects, fixed after the study; the release reports nothing



Dear ImGui 2 programs · 9d9a621f

frozen		**100%** 40 / 0 / 0
observed		**100%** 40 / 0 / 0
release		**100%** 40 / 0 / 0

1 true cause

- ● swapchain images transitioned before they are acquired



vkcube 1 program · 1.3.275

frozen		**100%** 2 / 0 / 0
observed		**100%** 2 / 0 / 0
release		**100%** 2 / 0 / 0

1 true cause

- ↑ present semaphore reused

Study 2: 25 programs, a later checker frozen before its first run



GTK 4 8 + 1 programs · 4.14.5

frozen		**14%** 173 / 0 / 1,048 / 13u
observed		**14%** 173 / 0 / 1,048 / 13u
release		**92%** 173 / 1 / 6 / 8u

3 true causes

- ↑ rendering never waits for the acquire semaphore
- ↑ present waits for acquire, never for rendering
- ↑ acquire semaphore re-signalled while a present may wait

false alarms: all of a frame's images in one descriptor array (frozen); atlas rectangles (6, release)



vkmark 7 scenes · ab6e6f3

frozen		**100%** 38 / 0 / 0
observed		**100%** 38 / 0 / 0
release		**100%** 38 / 0 / 0

5 true causes

- ↑ host rewrites the uniform buffer in flight
- ↑ acquire semaphore reused
- ● present semaphore reused
- ● clear barrier does not chain to the acquire wait
- ● depth image shared across frames, colour-only dependency



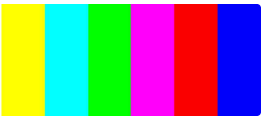
mpv / libplacebo 4 configurations · 0.37 / 6.338

frozen		**100%** 16 / 0 / 0
observed		**100%** 16 / 0 / 0
release		**100%** 16 / 0 / 0

1 true cause

- ↑ barrier does not chain to an external semaphore wait

frozen: timestamp queries unmodelled (no memory effect), outside the premises



GStreamer 5 pipelines · 1.24.2

frozen		**100%** 33 / 0 / 0
observed		**100%** 33 / 0 / 0
release		**100%** 33 / 0 / 0

2 true causes

- ● acquire wait at BOTTOM_OF_PIPE blocks nothing
- ● sampled image visible to input-attachment reads only

true · technicality · unresolved · false alarm

frozen: the fresh result (conservative capture, the theorem's configuration) · observed: same checker, observed capture · *release: post-hoc, same traces* · counts: true / technicality / false alarm (/ unresolved)

↑ changed upstream before the study ● still present upstream